Assignment 3
(Navigation & UI Architecture)

Team Members

1. Mariam Ashraf Hassan Mohamed
2. Hager Essam Gaber Mahmoud
3. Rama Ehab Ahmed Said
4. Seif Ashraf Fawzy Roushdy

1. Navigation Implementation Strategy

In alignment with the assignment requirements, we successfully implemented a multi-screen navigation system. Although the assignment allowed using any navigator, we opted for **State-Based Conditional Navigation** to ensure maximum security and performance for a cyber-security dashboard.

- **Total Screens Built:** 6 Screens (Login, Main Dashboard, Intelligence Center (Hub), Static Rules Engine, Anomaly Detection, and Logs Archive).
- **Default Screen:** The `LoginPage` was set as the default entry point. Access to other screens is strictly restricted until successful authentication.

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2. Addressing Assignment Concepts

We integrated the core concepts requested by the assignment into our modular architecture:

- **How to set default Screen:** Implemented in `App.js` using a `userToken` state. If null, the system defaults to the `LoginPage`.
- **Nested Navigation Logic:** While we used a flat state manager for performance, we simulated **Nested Navigation** within the `Intelligence Center (Hub)`. The Hub acts as a "Parent Component" that navigates users into specific "Child Modules" (Static, Anomaly, and Threat views).
- **Screen Options & Customization:** Every screen was customized with unique properties including:
  - **Dynamic Theming:** Full support for Dark/Light mode using Tailwind CSS.
  - **Access Control:** Conditional rendering based on user roles (Admin vs. Analyst).
  - **Transition Effects:** Smooth CSS transitions for screen switching.

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3. Documented Development Steps

1- Architecture Setup: Designed a central state controller in `App.js` to manage the `view` state across the application.
2- Component Decoupling: Separated the UI into modular components (Dashboard, DetectionHub, LogsHistory) to ensure scalability.

3- Authentication Flow: Created a secure login flow that communicates with the FastAPI backend and stores tokens in `localStorage`.
4- The Intelligence Hub: Developed a central "Navigation Hub" featuring high-end UI cards with Glassmorphism effects to route users to advanced AI detection modules.
5- Global Theme Management: Integrated a `ThemeToggle` component that updates the global CSS context, ensuring consistent branding across all navigation stacks.

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4. Screenshots

- Implementation of State-Based Conditional Rendering as an alternative to traditional navigators. This approach ensures centralized access control and optimized performance for a modular security dashboard.

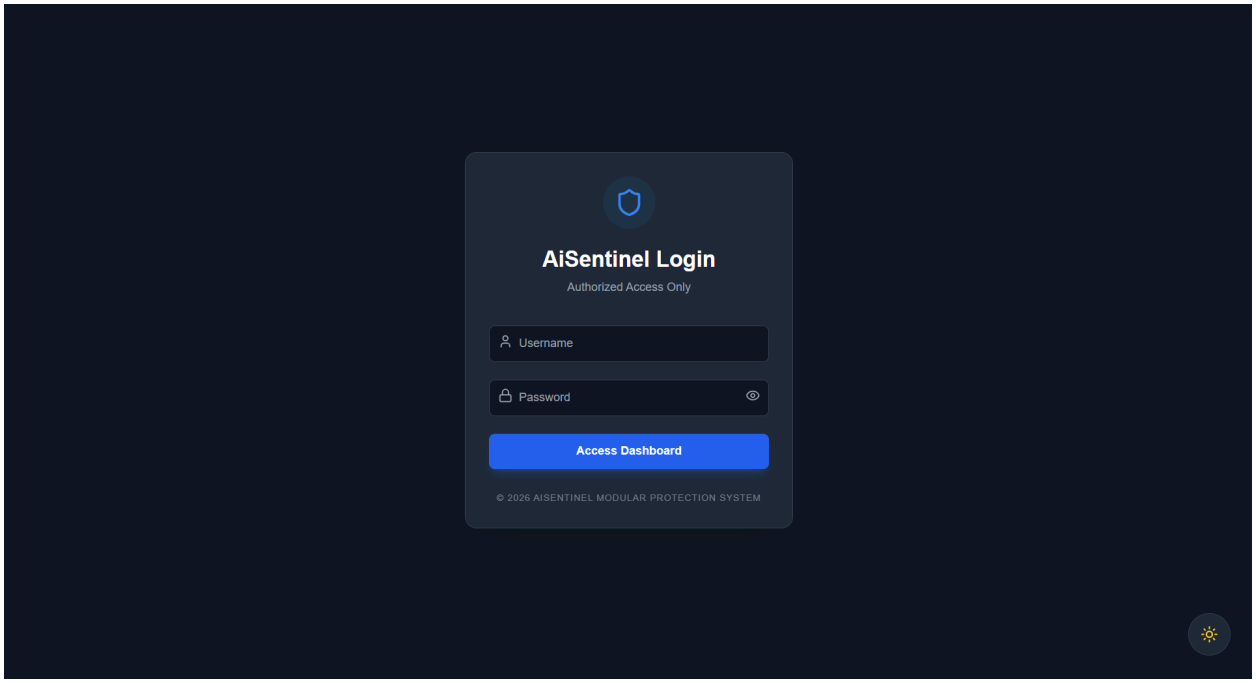
```
return (
<div className={darkMode ? "dark" : ""}>
  <div className="min-h-screen bg-gray-50 dark:bg-gray-900 flex flex-col transition-colors duration-300">

    <div className="flex-grow">
      {view === 'dashboard' ? (
        <Dashboard
          onGoToArchive={() => setView('archive')}
          onGoToHub={() => setView('hub')}
          onGoToUsers={() => setView('users')}
          onLogout={handleLogout}
        />
      ) : view === 'hub' ? (
        <DetectionHub
          onNavigate={(target) => setView(target)}
          onBack={() => setView('dashboard')}
        />
      ) : view === 'archive' ? (
        <LogsHistory onBack={() => setView('dashboard')} />
      ) : view === 'users' ? (
        <UserManagement onBack={() => setView('dashboard')} />
      ) : (
        <div className="p-8">
          <button onClick={() => setView('hub')} className="text-blue-500 mb-4">Back to Hub</button>
          <h2 className="text-2xl text-white uppercase font-black">{view} Analysis View</h2>
          <p className="text-gray-400 mt-2">Detailed results for {view} will appear here.</p>
        </div>
      )}
    </div>

    <ThemeToggle darkMode={darkMode} setDarkMode={setDarkMode} />

    <footer className="p-8 text-center text-gray-500 text-xs font-medium uppercase tracking-widest transition-colors">
      &copy; 2026 AiSentinel Modular AI System | Adaptive Data Protection
    </footer>
  </div>
</div>
);
```

- Our Screens


AiSentinel Login
Authorized Access Only
Username
Password
Access Dashboard
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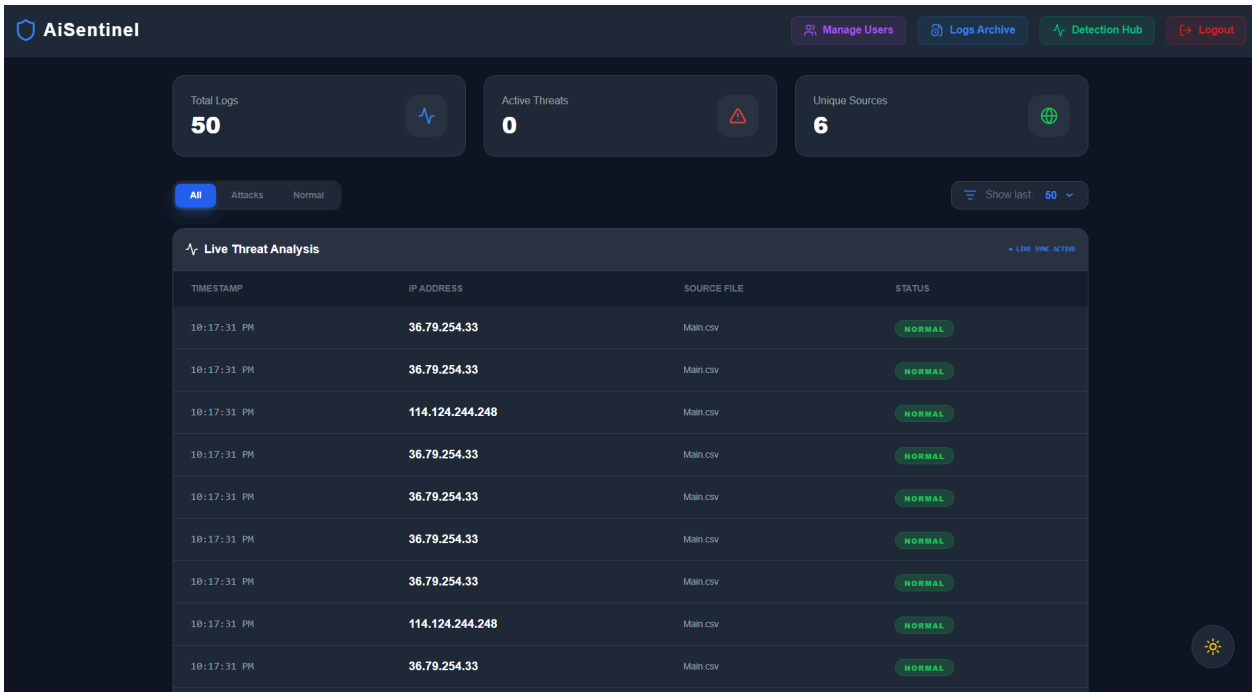
AiSentinel
Manage Users
Logs Archive
Detection Hub
Logout
Total Logs
50
Active Threats
0
Unique Sources
6
All
Attacks
Normal
Show last: 50
Live Threat Analysis
LIVE SYNC ACTIVE
TIMESTAMP
IP ADDRESS
SOURCE FILE
STATUS
10:17:31 PM 36.79.254.33 Main.csv NORMAL
10:17:31 PM 36.79.254.33 Main.csv NORMAL
10:17:31 PM 114.124.244.248 Main.csv NORMAL
10:17:31 PM 36.79.254.33 Main.csv NORMAL
10:17:31 PM 36.79.254.33 Main.csv NORMAL
10:17:31 PM 36.79.254.33 Main.csv NORMAL
10:17:31 PM 36.79.254.33 Main.csv NORMAL
10:17:31 PM 114.124.244.248 Main.csv NORMAL
10:17:31 PM 36.79.254.33 Main.csv NORMAL

USER MANAGEMENT

Add New User

USERNAME	ROLE	STATUS	ACTIONS
admin_boss	SUPER_ADMIN	ACTIVE	
analyst_1	ANALYST	ACTIVE	
analyst_2	ANALYST	PENDING UPDATE	
analyst_3	ANALYST	ACTIVE	
analyst_4	ANALYST	ACTIVE	
analyst_6	ANALYST	PENDING UPDATE	
analyst_7	ANALYST	PENDING UPDATE	

LOGS ARCHIVE

Search IP, 'attack' or 'normal'...

IP ADDRESS	TIMESTAMP	STATUS
36.79.254.33	2026-03-10T20:17:31.487Z	NORMAL
36.79.254.33	2026-03-10T20:17:31.487Z	NORMAL
114.124.244.248	2026-03-10T20:17:31.487Z	NORMAL
36.79.254.33	2026-03-10T20:17:31.487Z	NORMAL
36.79.254.33	2026-03-10T20:17:31.487Z	NORMAL
36.79.254.33	2026-03-10T20:17:31.487Z	NORMAL
36.79.254.33	2026-03-10T20:17:31.487Z	NORMAL
114.124.244.248	2026-03-10T20:17:31.487Z	NORMAL
36.79.254.33	2026-03-10T20:17:31.487Z	NORMAL
36.79.254.33	2026-03-10T20:17:31.486Z	NORMAL
114.124.244.248	2026-03-10T20:17:31.487Z	NORMAL
114.124.245.232	2026-03-10T20:17:31.486Z	NORMAL
36.81.234.203	2026-03-10T20:17:31.486Z	NORMAL
114.124.244.248	[illegible]	[illegible]

DETECTION HUB

Static Rules Engine

Traditional signature-based detection and firewall rules.

EXPLORE MODULE →

Anomaly Detection

AI-powered behavioral analysis to find unusual patterns.

EXPLORE MODULE →

Threat Detection

Real-time identification of known attack vectors (DDoS, Starvation).

EXPLORE MODULE →

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